SalaryScope

Model Analytics Report

Generated on: 2026-03-31 21:31:10

Performance Metrics

Test R^2 (log scale): 0.5949

Mean Absolute Error (MAE): $35,913.10

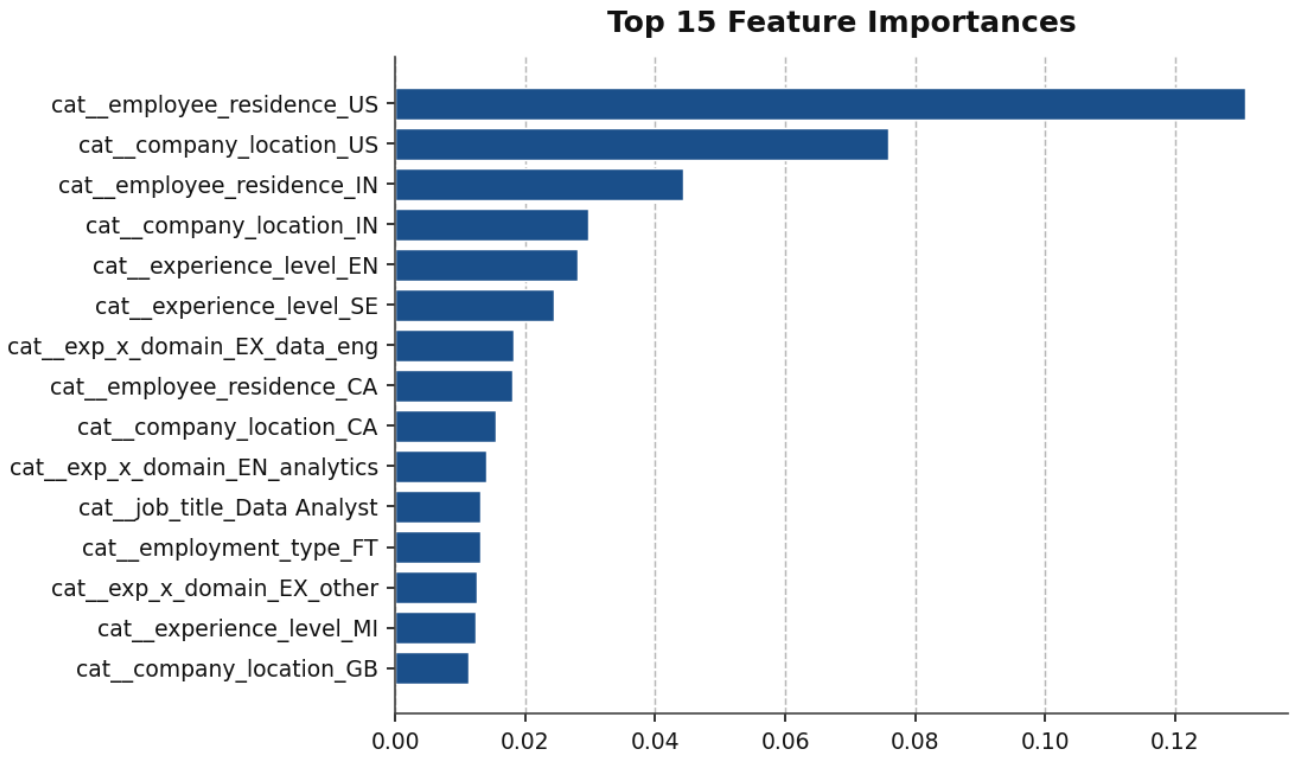
Root Mean Squared Error (RMSE): $48,774.45

Interpretation: The model explains approximately 59.5% of the variation in salary outcomes on the transformed scale. MAE reflects the average prediction error in USD, while RMSE places greater emphasis on larger errors.

Model Comparison

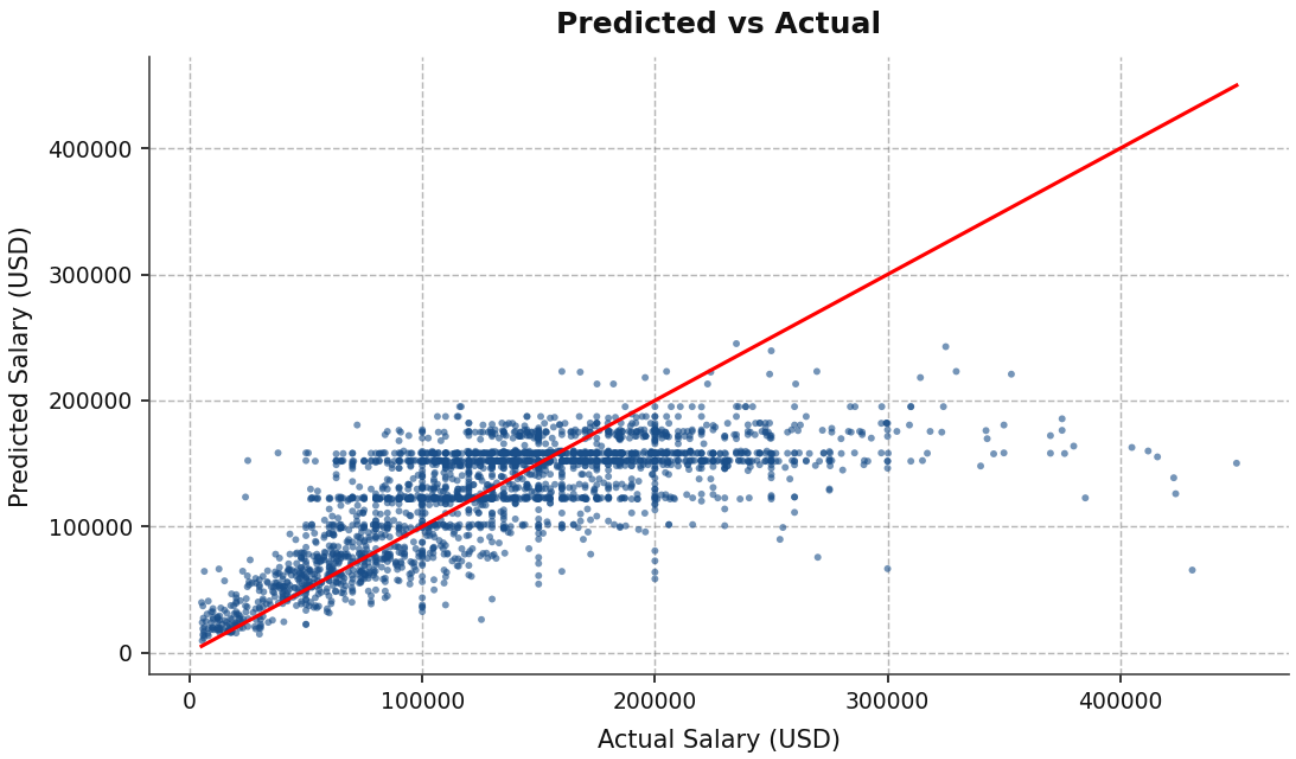
Model	R^2	MAE	RMSE
XGBoost (Raw + Engineered)	0.5949	$35,913	$48,774
XGBoost (Log)	0.5944	$37,668	$51,505
Random Forest (Log)	0.5759	$37,878	$51,768
Gradient Boosting (Raw)	0.3989	$38,921	$51,278
Linear Regression (Raw)	0.3486	$40,169	$53,380

Feature Importance



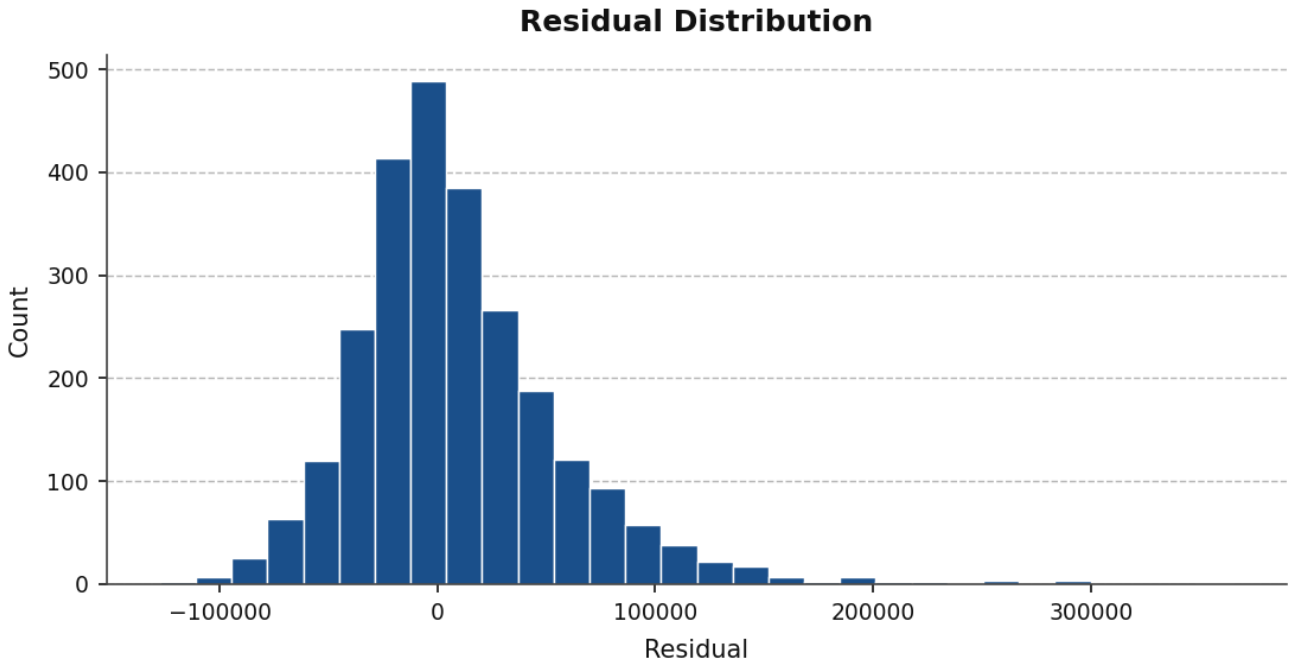
Interpretation: Feature importance shows which transformed variables contribute most strongly to salary prediction.

Predicted vs Actual Analysis



Interpretation: Each point represents one test observation. Points closer to the diagonal reference line indicate more accurate predictions.

Residual Analysis



Interpretation: Residuals represent the difference between actual and predicted salaries. A distribution centered near zero suggests balanced error behavior. Shifts away from zero may indicate slight underprediction.